

CS358 Midterm Project

Games with Greenfoot: Implementing Event Handling

Note: You will work in an assigned group (see Blackboard for your group assignment).

Step 1: Install Greenfoot

<https://www.greenfoot.org/download>

It runs on both Windows and Mac.

Step 2: Check out the example games

This project is intentionally open-ended. The following sample projects from class are provided as reference:

- **Hedgehog**
- **Zelda3** (move between screens, basic animation, simple enemies, simple dialog window)

Step 3: Build a game! Here are some ideas:

Pong



Implement two paddles controlled by different keyboard keys. The ball should bounce off the boundaries and paddles. If the ball passes a paddle, award a point to the opposing player.

Space Invaders



Move the player's ship along the bottom of the screen and fire projectiles at enemies. Enemies should advance toward the player over time.

Your game should be different than the examples

Your game should be meaningfully different from the provided examples. Avoid minor modifications; aim to implement original mechanics, assets, or gameplay features.

How to use Custom Images

Review the Zelda3 example and the provided demo video from its creator. It includes custom images stored in the project's **images** folder.

To add images, place them in the **images** folder within your Greenfoot project using your file manager (Finder on macOS or File Explorer on Windows).

To use a custom image in `YourActor`, add code similar to the following in the constructor:

```
public YourActor() {  
    GreenfootImage image = new GreenfootImage("images/yourfile.png");  
    setImage(image);  
}
```

Where to get new images

You may find new art assets on itch.io, which hosts indie games and downloadable asset packs. In game development, the term *assets* commonly refer to artwork such as buttons, icons, sprites (characters), and similar resources.

If you use third-party assets, you must provide appropriate credit to the creator. Keep a record of the assets you download (for example, by saving the itch.io link for each asset used).

Note: Crediting creators is required and is a professional best practice.

Step 4: Submit your project

- Compress (zip) your project folder and upload it to Blackboard.
- Include a write-up (word file or pdf) describing three interesting or technically challenging features your team implemented. This may be brief.
- In the same write-up, please answer the following questions (your responses will not impact on your midterm project grade; they will be used solely to evaluate the effectiveness of this assignment in teaching event-driven programming):
 1. How did you like the "unplugged" event-based activity we did in class, where you and your team had to draw and explain an overly complicated machine with five events? Did it help you better understand the fundamentals of event-driven programming?
 2. How do you like the event-driven examples in Greenfoot? Did they help you understand how event handling is implemented?
 3. How did you like developing your own Greenfoot game using event-based programming for this midterm project?
 - What was your favorite part?
 - Do you have any suggestions for improvement?

Grading Criteria:

Criteria	Exceptional (90-100%)	Satisfactory (70-89%)	Needs Improvement (0-69%)
Event-Driven Implementation	Successfully implements complex event handling (e.g., keyboard input, collisions) across multiple actors.	Implements basic event handling as seen in class examples.	Event handling is broken, missing, or purely copied from provided samples.
Originality & Complexity	Game is "meaningfully different" with original mechanics, assets, or gameplay features.	Game is functional but relies heavily on the mechanics of the Hedgehog, Zelda, or Pong examples.	Game is a "minor modification" of a class example.
Asset Management	Successfully integrates custom images/sprites using the images folder and proper code constructors.	Uses custom images, but with minor scaling or implementation issues.	Uses only default Greenfoot assets or fails to implement setImage() correctly.
Technical Write-up	Provides a clear description of three interesting or technically challenging features.	Describes features, but they lack technical detail or complexity.	Write-up is missing or fails to describe three distinct features.
Professionalism (Citations)	All third-party assets from itch.io or other sources are clearly credited with links.	Most assets are credited, but some links or creator names are missing.	No credit is given for third-party assets.